Project Development Phase: Performance Test

SkillWallet | Data Analytics with Tableau Capstone

Project detail	Confirmed information
Documentation date	27 October 2026
Final submission date	28 October 2026
Team ID	LMP-EV-2026
Project name	Electric Vehicle Market Performance & Insights
Tableau workbook	Electric Vehicle Market & Technology Analysis
Project owner	Yuvraj Shrivastava
Enrollment number	EN25CS3011105
Institution	Medicaps University, Indore
Contribution	Individual project completed by Yuvraj Shrivastava

Project identification

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Verified CSV source-file statistics

CSV inspection item	Verified value
Uploaded file	ev_market_train(1).csv
Rows / records	200,000
Columns	37
File size	46,478,548 bytes (44.33 MiB)
Null cells in uploaded CSV	18,996
Rows containing at least one null	18,251
Exact duplicate rows	1
Repeated vehicle_id values	1
Test CSV usage	Not connected to Tableau workbook, per project owner

Performance testing results

S.No.	Parameter	Updated screenshot / values
1.	Data Rendered	Connected source file: ev_market_train(1).csv. Verified source-file dimensions: 200,000 records × 37 columns. File size: 44.33 MiB (46,478,548 bytes). Actual uploaded schema is listed in the Data Preparation note. The separate test CSV was not connected to the workbook.

S.No.	Parameter	Updated screenshot / values
2.	Data Preprocessing	Important data-quality finding: direct inspection of the uploaded CSV shows 18,996 null cells across 18,251 rows, 1 exact duplicate row, and 1 repeated vehicle_id value. This means the uploaded file itself is not a fully cleaned export. You previously confirmed that nulls and duplicates were removed and data types corrected during preparation. The documents distinguish the supplied source-file inspection from the reported cleaning step; a cleaned final CSV or Tableau cleaning screenshot is needed to verify the post-cleaning result. No cleaned row count is invented. Confirmed process reported by owner: remove nulls, remove duplicates, correct data types. Post-cleaning totals are not verifiable from this source file alone.
3.	Utilization of Filters	Dashboard filters: model_year, category, battery_type and manufacturing_country. “Use as Filter” actions support cross-filtering. Filter response times and controlled performance comparisons were not recorded.
4.	Calculation fields Used	SUM(annual_sales_units); COUNTD(vehicle_id); AVG(range_km); AVG(price_usd); AVG(customer_rating); MEDIAN(range_km); Percent of Total. AVG(battery_capacity_kwh) is used in the rating/battery view.
5.	Dashboard design	9 Tableau worksheets and 1 interactive dashboard titled “EV Market Performance & Strategy Dashboard”. Views include KPI summary, category sales, manufacturer treemap, battery performance, battery share, manufacturing map, price-range scatter, rating/battery combination and country-battery matrix.
6.	Story Design	Story title: “Strategic Insights: Global EV Market & Battery Technology”. Only Story Point 1 is completed: “Global EV Market & OEM Overview”, using Category Sales and Manufacturer Share treemap. Story Points 2–4 are not completed.

Data quality clarification

Important data-quality finding: direct inspection of the uploaded CSV shows 18,996 null cells across 18,251 rows, 1 exact duplicate row, and 1 repeated vehicle_id value. This means the uploaded file itself is not a fully cleaned export. You previously confirmed that nulls and duplicates were removed and data types corrected during preparation. The documents distinguish the supplied source-file inspection from the reported cleaning step; a cleaned final CSV or Tableau cleaning screenshot is needed to verify the post-cleaning result. No cleaned row count is invented.

Actual dataset columns

Field group	Actual columns in uploaded CSV
Identifiers / segments	vehicle_id, model_name, manufacturer, model_year, manufacturing_country, market_region, category
Battery / technical	battery_capacity_kwh, battery_type, battery_cycle_life, battery_degradation_pct_5yr, range_km, efficiency_wh_per_km, motor_power_kw, torque_nm, num_motors, drivetrain, top_speed_kmph, acceleration_0_100_sec, max_dc_charging_kw, charging_dc_fast_time_min, charging_ac_time_hr, num_charging_ports, num_seats, weight_kg, cargo_capacity_liters
Commercial / customer	price_usd, insurance_cost_usd_yearly, warranty_years, resale_value_pct_3yr, safety_rating_stars, customer_rating, annual_sales_units
Sustainability / usage	co2_saved_tons_per_year, autonomous_level, infotainment_screen_inch, recommended_use_case

Tableau publication references

Tableau Public workbook: https://public.tableau.com/views/ElectricVechileMarketTechnologyAnalysis/ElectricVehicleMarketPerformanceInsights?:showVizHome=no

Live project website: https://en25cs3011105-spec.github.io/Electric-Vehicle-Market-Performance-and-Insights/

Testing limitations

The supplied screenshots establish that the Tableau views render. No measured filter timings, workbook load duration, browser response-time tests, Tableau query-history export or cleaned output CSV were supplied. Such performance figures are not asserted.